

Generation Interconnection System Impact Study Report

For Queue Project Q21-1770

Amber 138 kV
February 22, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1770. The Interconnection Customer proposes a 249.7 MW solar photovoltaic facility interconnecting at the Amber 138 kV substation in the Virginia area. The request was received on June 06, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 472 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 16 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$104,607,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Amber – Bremono Bluff 138 KV Line	Lake Lure – Bluff City 138 KV	181.6	194	45.8
Bremono Bluff – Willow Crk 138 KV line	Louisa Generation Facility – Fluvanna P.S 345 KV line	142.8	245	6.0
Willow Creek – Golden Hills 138 KV	Louisa Generation Facility – Fluvanna P.S 345 KV	140.9	240	6.0
Instituto Tecnologico De Ponce – Golden Hills 138 KV	Louisa Generation Facility Substation – Fluvanna P.S Substation 345 KV Line	138.7	238	6.0
Instituto Tecnologico De Ponce – Abbeville 138 KV line	Fluvanna P.S – Louisa Generation Facility 345 KV	136.8	236	6.0
Abbeville – Cabrillo 138 KV L	Louisa Generation Facility – Fluvanna P.S 345 KV	134.8	235	6.0
Cabrillo Substation – Fern Vly 138 KV	Fluvanna P.S Substation – Louisa Generation Facility 345 KV L	134.7	227	6.0
Fern Vly – Apple Data Ctr Pv 138 KV	Fluvanna P.S – Louisa Generation Facility Substation 345 KV Line	134.4	223	6.0
Apple Data Ctr Pv – Cold Spring Rd 138 KV	Fluvanna P.S Substation – Louisa Generation Facility 345 KV	131.6	210	6.0
Cold Spring Rd – N Anna 138 KV Line	Fluvanna P.S – Louisa Generation Facility 345 KV	131.1	211	6.0
Ellinwod – Franklin-Laguna 138 KV	Fluvanna P.S – Louisa Generation Facility 345 KV	114.5	262	5.8
Kell Butte – Lake Lure Substation 138 KV	Lake Lure – Littlefork 138 KV	113.1	228	7.2
Orrington – Unisource Golden Vly 138 KV	Louisa Generation Facility – Fluvanna P.S 345 KV	110.6	250	5.8

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Kell Butte – Unisource Golden Vly 138 KV Line	Lake Lure – Littlefork Substation 138 KV	108.3	212	7.2
Orrington – Ellinwood 138 KV	Fluvanna P.S – Louisa Generation Facility 345 KV	102.8	285	5.8
Dunn Siding – Bear Island Substation 345 KV	Leavitt – Birchwod 345 KV line	100.3	2,769	8.2

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Amber – Bremono Bluff 138 KV Line	Rebuild line with higher-capacity conductor	\$13,991,000	100% (\$13,991,000)
Bremono Bluff – Willow Crk 138 KV line	Rebuild line with higher-capacity conductor	\$5,300,000	100% (\$5,300,000)
Willow Creek – Golden Hills 138 KV	Rebuild line with higher-capacity conductor	\$6,886,000	100% (\$6,886,000)
Instituto Tecnologico De Ponce – Golden Hills 138 KV	Rebuild line with higher-capacity conductor	\$6,428,000	100% (\$6,428,000)
Instituto Tecnologico De Ponce – Abbeville 138 KV line	Rebuild line with higher-capacity conductor	\$4,638,000	100% (\$4,638,000)
Abbeville – Cabrillo 138 KV L	Rebuild line with higher-capacity conductor	\$4,629,000	100% (\$4,629,000)
Fern Valley – Cabrillo 138 KV	Rebuild line with higher-capacity conductor	\$12,058,000	100% (\$12,058,000)
Fern Vly – Apple Data Ctr Pv 138 KV	Rebuild line with higher-capacity conductor	\$8,566,000	100% (\$8,566,000)
Apple Data Ctr Pv – Cold Spring Rd 138 KV	Rebuild line with higher-capacity conductor	\$4,530,000	100% (\$4,530,000)
N Anna – Cold Spring Rd Substation 138 KV	Rebuild line with higher-capacity conductor	\$2,329,000	100% (\$2,329,000)
Ellinwod – Franklin-Laguna 138 KV	Rebuild line with higher-capacity conductor	\$18,418,000	100% (\$18,418,000)
Kell Butte – Lake Lure 138 KV line	Rebuild line with higher-capacity conductor	\$6,565,000	82% (\$5,418,000)
Orrington – Unisource Golden Vly 138 KV	Rebuild line with higher-capacity conductor	\$2,663,000	100% (\$2,663,000)
Kell Butte – Unisource Golden Vly 138 KV Line	Reconductor line	\$7,569,000	82% (\$6,247,000)
Orrington – Ellinwood 138 KV	Reconductor line	\$2,425,000	100% (\$2,425,000)
Bear Island – Dunn Siding 345 KV	Reconductor line	\$13,577,000	1% (\$81,000)
	Total allocated to project		\$104,607,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.